Kalvik Jakkala

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Professional Summary

Research scientist specializing at the intersection of machine learning and field robotics. Backed by extensive hands-on experience, with a proven track record of translating complex algorithms into field-ready solutions and deploying physical robots in dynamic, real-world environments.

Core Expertise:

- **Machine Learning:** Bayesian learning theory, Gaussian & Sparse Gaussian processes, real-time applications.
- **Sensing & Navigation:** Multi-modal sensing (camera, sonar, radar), robust autonomous navigation systems.
- **Hardware & Deployment:** Real-world robotic deployment with resource-constrained edge devices.

Research Experience

Postdoctoral Researcher, Texas A&M University **Oct 2024—Present**
Field Robotics and Autonomous Navigation
Advisor: Dr. Jason O'Kane

- **Algorithm Design:** Developing autonomous navigation algorithms for underwater robots operating under strict compute constraints, zero reliance on global localization, and minimal sensing.
- **Hardware Deployment:** Successfully optimizing and deploying computationally efficient methods onto physical robots utilizing edge devices, including Nvidia Jetson and Raspberry Pi platforms.

Doctoral Researcher, University of North Carolina at Charlotte **May 2020—Sep 2024**
Bayesian Sensor Placement and Informative Path Planning

- **Software Engineering:** Developed efficient methods for sensor placement and informative path planning in continuous and discrete domains, notably developing the `SGP-Tools` Python library.
- **Theoretical Foundations:** Established guarantees for estimation uncertainty under motion constraints to ensure robust, predictable performance during autonomous data collection.
- **Performance Optimization:** Achieved processing speeds an order of magnitude faster than prior state-of-the-art solutions, enabling real-time deployment on resource-constrained field robots in marine environments.

Research Intern, GE Research **May 2023—Aug 2023**

- **Explainable AI:** Created a low size, weight, and power-optimized Bayesian framework to explain and justify the predictions of pre-trained deep neural networks.

Graduate Researcher, University of North Carolina at Charlotte **Jan 2018—May 2020**
Applied Deep Learning

- **Radar Pose Estimation (2019—2020):** Designed and evaluated deep learning models for human pose estimation and activity recognition using mmWave radar data.
- **Signal Processing (2018—2020):** Developed deep learning algorithms for gait-based user identification leveraging signals from WiFi routers and mmWave radars.

Selected Publications

[1] **Kalvik Jakkala**, Saurav Agarwal, Jason O'Kane, and Srinivas Akella. "Informative Path Planning with Guaranteed Estimation Uncertainty". In: *Robotics: Science and Systems (RSS)*. 2026. URL: https://www.itskalvik.com/publication/uncertainty-guaranteed-ipp/.

[2] **Kalvik Jakkala**, Saurav Agarwal, Jason O'Kane, and Srinivas Akella. "Robot-Motion-Uncertainty-Aware Informative Path Planning with Guaranteed Estimation Uncertainty". Under Review. 2026.

[3] **Kalvik Jakkala** and Jason O'Kane. "Low-Cost Underwater In-Pipe Centering and Inspection Using a Minimal-Sensing Robot". Under Review. 2026. URL: https://www.itskalvik.com/publication/underwater-in-pipe-centering/.

[4] **Kalvik Jakkala** and Jason O'Kane. "Schur-MI: Fast Mutual Information for Robotic Information Gathering". In: *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE/RSJ, 2026. URL: https://www.itskalvik.com/publication/schur-mi/.

[5] **Kalvik Jakkala** and Srinivas Akella. "Fully Differentiable Adaptive Informative Path Planning". In: *IEEE International Conference on Robotics and Automation, ICRA*. May 2025, pp. 5431–5437. URL: https://www.itskalvik.com/publication/sgp-aipp/.

[6] **Kalvik Jakkala** and Srinivas Akella. "Fully differentiable sensor placement and informative path planning". In: *The International Journal of Robotics Research (IJRR)* (2025). URL: https://www.itskalvik.com/publication/sgp-foundation/.

[7] **Kalvik Jakkala** and Srinivas Akella. "Multi-Robot Informative Path Planning from Regression with Sparse Gaussian Processes". In: *IEEE International Conference on Robotics and Automation, ICRA*. IEEE, 2024. URL: https://www.itskalvik.com/publication/sgp-ipp/.

[8] **Kalvik Jakkala** and Srinivas Akella. "Probabilistic Gas Leak Rate Estimation Using Submodular Function Maximization With Routing Constraints". In: *IEEE Robotics and Automation Letters, RA-L* (2022). URL: https://www.itskalvik.com/publication/graph-ipp/.

Education

Ph.D. in Computer Science, University of North Carolina at Charlotte — **Aug 2018—May 2024**

- **Concentration:** Machine Learning and Robotics **Advisor:** Dr. Srinivas Akella
- **Thesis:** Efficient Bayesian Sensor Placement and Informative Path Planning

M.Sc. in Computer Science, University of North Carolina at Charlotte — **Aug 2018—May 2021**

- **Concentration:** Artificial Intelligence, Robotics, and Gaming **GPA:** 4.00 / 4.00

B.Sc. in Computer Science, Wichita State University — **Aug 2014—May 2018**

- **Minor:** Mathematics **GPA:** 3.45 / 4.00

Teaching Experience

Teaching Assistant, University of North Carolina at Charlotte — **Jan 2021—May 2024**
Mentored graduate students in: Machine Learning (ITCS 8156), Algorithms and Data Structures (ITCS 8114), and Optimization for Machine Learning and Data Science (ITCS 8010).

B.S. Teaching Fellow, Wichita State University — **Aug 2016—May 2018**
Co-instructed and tutored undergraduate students in: Object-Oriented Programming (CS 311), Data Structures (CS 300), and Introductory C++ Programming (CS 211).

Service & Awards

- **Associate Editor:** IEEE International Conference on Robotics and Automation (ICRA) — **Sep 2025—Present**
- **Co-organizer:** IROS 2026 Workshop on Perception and Navigation for Marine Robotics — **Sep 2026**
- **Reviewer:** Top-tier robotics journals/conferences (ICRA, IROS, IJRR, RA-L, MRS, Ocean Eng.) — **Jan 2022—Present**
- **Grant Recipient:** UNC Charlotte Graduate School's Summer Fellowship (GSSF) grant — **May 2022**
- **Deans Honor Roll:** Recognized for outstanding academic performance — **2014—2018**
- **Vice President:** Association for Computing Machinery (ACM) — **Aug 2015—Dec 2016**
- **Vice President:** Institute of Electrical and Electronics Engineering (IEEE) — **Aug 2015—Dec 2016**

Skills

- **Hardware & Edge Compute:** Nvidia Jetson, Raspberry Pi | **Sensors & Perception:** Sonar, mmWave Radar, Vision/Cameras
- **Programming Languages:** Python, C/C++, Matlab, Bash | **Libraries:** TensorFlow, PyTorch, OpenCV, ROS, GPflow, JAX
- **Platforms & Tools:** Linux, Docker, Slurm, Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- **Domain Expertise:** Field Robotics Deployment (Autonomous Underwater/Surface/Ground/Aerial Vehicles)